

Sample Data and Test Cases

1. Inventory Reconciliation Test Cases

Test Case 1

Input

Inventory.csv

asset_id,hostname,ip_address

A001,Server01,192.168.1.10

A002,Server02,192.168.1.11

A003,Server03,192.168.1.12

LiveInventory.json

```
[
{
  "asset_id": "A001",
  "hostname": "Server01",
  "ip_address": "192.168.1.10"
},
{
  "asset_id": "A003",
  "hostname": "Server03",
  "ip_address": "192.168.1.12"
}
]
```

Expected Output

```
{
  "matched_assets": 2,
  "missing_assets": 1,
  "untracked_assets": 0
}
```

Test Case 2

Input

Inventory.csv

asset_id,hostname

A001,Server01

A002,Server02

LiveInventory.json

```
[  
  {  
    "asset_id": "A001",  
    "hostname": "Server01"  
  },  
  {  
    "asset_id": "A005",  
    "hostname": "UnknownServer"  
  }  
]
```

Expected Output

```
{  
  "matched_assets": 1,  
  "missing_assets": 1,  
  "untracked_assets": 1  
}
```

2. Inventory Validation Agent Test Cases

Valid File

Input

asset_id,hostname

A001,Server01

A002,Server02

Expected Output

```
{  
  "status": "valid"  
}
```

Invalid File

Input

asset_id,hostname

A001,

A001,

Expected Output

```
{  
  "duplicates_detected": true,  
  "missing_values": true  
}
```

3. Reconciliation Agent Test Cases

Naming Mismatch

Input

Inventory

DB-SERVER-01

Live

db_server_01

Expected Output

```
{
```

```
"naming_mismatch": true
}
```

Configuration Mismatch

Input

Inventory IP

192.168.1.10

Live IP

192.168.1.50

Expected Output

```
{
  "configuration_mismatch": true
}
```

4. AI Analysis Agent Test Cases

Input

```
{
  "issue": "Missing Asset",
  "asset": "Server02"
}
```

Expected Output

```
{
  "root_cause": "Inventory not updated",
  "risk": "Medium"
}
```

Input

```
{
```

```
"issue":"Untracked Asset",  
"asset":"UnknownServer01"  
}
```

Expected Output

```
{  
  "root_cause":"Unauthorized asset or inventory gap",  
  "risk":"High"  
}
```

5. AI Chatbot Test Cases

Question

Show missing assets.

Expected Response

Server02

Server06

Question

Summarize inventory health.

Expected Response

Inventory reconciliation completed.

Missing Assets: 2

Untracked Assets: 1

Overall Status:

Attention Required

6. Report Generation Test Case

Input

```
{  
  "total_assets":6,  
  "missing_assets":2,  
  "untracked_assets":1  
}
```

Expected Result

Generated PDF containing:

- Reconciliation Summary
- Missing Assets
- Untracked Assets
- AI Analysis
- Recommendations
- Executive Summary
- Timestamp

Demo Scenario for Review

Login

Admin

admin@inventoryai.com

Password@123

Infrastructure Engineer

engineer@inventoryai.com

Engineer@123

Auditor

auditor@inventoryai.com

Auditor@123

System Flow

Infrastructure Engineer Login



Upload Inventory.csv



Upload LiveInventory.json



Inventory Validation Agent



Reconciliation Agent



AI Analysis Agent



Recommendation Agent



Executive Summary Agent



Report Generation Agent



AI Chatbot

Output

Matched Assets

Missing Assets

Untracked Assets

Configuration Mismatches

Naming Mismatches

AI Analysis

[Recommendations](#)

[Executive Summary](#)

[Downloadable PDF Report](#)

[Chatbot Responses](#)

This provides a complete set of realistic sample data, test cases, and demonstration scenarios for documentation, testing, project reviews, and viva presentations.